One-Hot-Preserving QAOA for Flood Shelter Assignment: Pairwise Capacity Encoding and Angle Transfer

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***Abstract*—Shelter assignment after a flood disruption is a small assignment problem with capacity inequalities that is often proposed for quantum optimization. We study the Quantum Approximate Optimization Algorithm (QAOA) on 12-qubit instances from a flood digital twin by exact statevector simulation. Capacity is encoded with pairwise penalties instead of slack qubits; the encoding is approximate, and its minimiser was the true optimum in 93.3% of instances. A ring-XY mixer that keeps each group assigned to one shelter placed a mean probability of 0.188 on the optimal plan at depth 4 over 5 optimizer seeds, against at most 0.048 for the transverse-field mixer at the same CNOT count per layer. One angle set per depth, chosen among the angles trained on 30 development instances, was applied unchanged to 30 held-out seeds of the same disruption pattern. Chosen by mean optimal-plan probability, which needs the optima of the development instances, it reached 0.306 against 0.163 for per-instance training. Chosen without optima, by mean energy or CVaR, it reached 0.264 and 0.234, but fell below per-instance training in leave-one-out tests. Transfer failed without a shared energy scale. On 50 instances on the road networks of 6 Indian cities, fixed angles left the mean probability unchanged but raised the chance that 32 verified samples contain the optimum. Exhaustive search solves every instance in about a millisecond; we claim no quantum advantage.**

***Index Terms*—QAOA, XY mixer, parameter transfer, QUBO, evacuation planning**

I. Introduction

When a flood closes roads and fills shelters, an emergency officer must decide again which shelter each evacuation group should go to, trading off travel time, flood exposure and the seats left. As a binary program this is a capacity-constrained assignment problem, the kind that is routinely encoded as a quadratic unconstrained binary optimization (QUBO) problem [1], [2] and offered as an application for near-term quantum optimization [3], [4].

Prior proposals rarely address three questions. First, does the circuit respect the constraints or only penalise violations? The original QAOA [5] explores all 2^n bitstrings, most of which are not plans, whereas the quantum alternating operator ansatz [6] keeps the state inside a feasible subspace [7]–[9]. Mixer studies treat equality constraints, while shelter capacity is an inequality. Second, must the angles be trained again after every disruption, and how should a reusable angle set be chosen? Third, how do the results compare with classical methods and with what a sampler plus a feasibility check returns?

We address them on instances generated by the digital twin of a flood decision-support prototype. Our contributions are:

1) A shelter-assignment QUBO in which capacity is encoded by pairwise products instead of slack qubits, with a cost shift and scaling that place all instances on one energy range; we measure how often the encoding is exact and show that it admits infeasible zero-penalty plans.
2) A comparison of transverse-field, ring-XY, Grover and feasible-subspace mixers under mean-energy and conditional value-at-risk (CVaR) training over 5 optimizer seeds, with Holm-corrected tests and CNOT counts per layer.
3) An angle-transfer study with three selection rules, two of which need no optima, an ablation in which transfer failed without the shared energy scale, and evidence that the performance of trained angle sets concentrates across instances.
4) The expected optimality gap of the best verified plan among N samples, and a comparison with exhaustive search, a greedy rule and simulated annealing.

All results come from classical simulation; we make no claim of quantum speed-up.

II. Related Work

Formulations. Lucas gave Ising forms for many NP-hard problems [1], and Glover *et al.* describe how assignment and capacity constraints become quadratic penalties, with slack variables for inequalities [2]. Flood shelter allocation has been solved by integer programming [10] and by quantum annealing [3].

Mixers. QAOA alternates a cost unitary and a mixing unitary for p layers [5], [11]. Hadfield *et al.* generalised the mixer so that it preserves hard constraints [6]. Wang *et al.* analysed ring and complete XY mixers for one-hot constraints [7], Fuchs *et al.* gave general constructions of constraint-preserving mixers [8] and mixers restricted to a subspace [12], and Grover mixers move the complexity into the preparation of the feasible superposition [9]. Applied studies pair XY mixers with warm starts for vehicle routing [13] and build coverage-preserving mixers for rostering [14].

Training and transfer. Angles can be interpolated from depth p to $p+1$ [15], CVaR objectives raise the probability of good samples [16], and warm starts use classical solutions [17]. The objective value at fixed angles concentrates across typical instances [18], optimal angles transfer between random

graphs [19], and for weighted MaxCut a rescaling of the weights lets one parameter set serve many instances [20]. On hardware, QAOA performance falls with size and depth [21], and practical advantage has not been shown [4].

Positioning. We do not propose a new mixer; the ring-XY mixer is that of [7]. The transfer studies [18]–[20] use unconstrained MaxCut with the standard mixer. We add a capacity inequality kept as a pairwise penalty next to a one-hot-preserving mixer, compare selection rules with and without optima, and evaluate on held-out instances and other road networks with a verifier in the loop.

III. Problem Formulation

A. Instances from the Digital Twin

The twin models a corridor with three population zones, nine junctions and three shelters joined by 22 roads. Every road carries a travel time, a flood-depth index and a congestion level drawn from a seed. Shelters have capacities of 120, 90 and 70 people with an initial occupancy of 15–45%, and groups have 8–22 people. Each instance applies the disruption used in the twin's routing benchmark: road R5 closes and Shelter A receives arrivals equal to half its capacity. For group g and shelter s we take the best open path under the twin's flood- and congestion-weighted edge cost and define

$$c_{gs} = T_{gs} + 5H_{gs}, \tag{1}$$

where T_{gs} is travel time in minutes and H_{gs} the mean flood index along the path. The pair (g, s) is *forbidden* if no open path exists, the shelter has failed, or the group alone exceeds the remaining seats R_s. With G groups, binary variables x_{gs} give $n = 3G$ qubits. We write m_g for the number of people in group g. A plan is feasible if every group has exactly one shelter, no forbidden pair is used and

$$\textstyle\sum_g m_g x_{gs} \le R_s \quad \text{for each shelter } s. \tag{2}$$

B. QUBO with Pairwise Capacity

An inequality normally needs slack qubits [2]. We use a pairwise encoding instead. Let $\mathcal{O} = \{(g, h, s) : m_g + m_h > R_s\}$ be the pairs of groups that cannot share shelter s and $\mathcal{F}$ the forbidden pairs. With shifted costs $c'_{gs} = c_{gs} - \min_{s'} c_{gs'}$ and penalty weight μ,

$$E(\mathbf{x}) = \sum_{g,s} c'_{gs} x_{gs} + \mu\Big[\sum_g \Big(1 - \sum_s x_{gs}\Big)^2 + \sum_{(g,s)\in\mathcal{F}} x_{gs} + \sum_{(g,h,s)\in\mathcal{O}} x_{gs}x_{hs}\Big]. \tag{3}$$

Remark 1: With $\mu = 1 + \sum_g \max_s c'_{gs}$, every bitstring that breaks a one-hot, forbidden-pair or pairwise-capacity term has higher energy than every bitstring that breaks none of them, since $c'_{gs} \ge 0$ bounds the cost of a bitstring without violations by $\mu - 1$ and each violation adds at least μ. The bound no longer holds for the reduced weights $\lambda\mu$ with $\lambda < 1$ used in the penalty sweep of Section VI-A.

On one-hot states the shift subtracts the constant $\sum_g \min_s c_{gs}$, so the optimal plan is unchanged. We divide the Hamiltonian by $\Delta = \mu - 1$, which maps the energies of all penalty-free plans into $[0, 1]$ for every instance. The pairwise encoding misses overloads by three or more groups whose pairs all fit, so a penalty-free plan can still be infeasible, and every sampled plan is re-checked exactly before use.

IV. QAOA Variants and Training

For depth p and angles $\boldsymbol{\theta} = (\boldsymbol{\gamma}, \boldsymbol{\beta})$, QAOA prepares

$$|\psi(\boldsymbol{\theta})\rangle = \prod_{k=1}^{p} e^{-i\beta_k B}\, e^{-i\gamma_k \hat{E}/\Delta}\, |\psi_0\rangle, \tag{4}$$

where $\hat{E}$ is diagonal with entries $E(\mathbf{x})$. Substituting $x_q = (1 - Z_q)/2$ turns (3) into an Ising Hamiltonian with one R_Z rotation per qubit and one ZZ rotation, two CNOTs, per non-zero coupling.

Transverse-field (X) mixer. $B = \sum_q X_q$ and $|\psi_0\rangle = |+\rangle^{\otimes n}$; constraints act only through the penalties. A cost layer has 12 one-hot couplings and on average 2.0 capacity couplings, 28 CNOTs; the mixer needs none.

Ring-XY mixer. For each group g with qubits a_g, b_g, c_g,

$$B_{\mathrm{XY}} = \sum_g \sum_{(u,v)\in\{(a_g,b_g),(b_g,c_g),(c_g,a_g)\}} \tfrac{1}{2}\big(X_u X_v + Y_u Y_v\big). \tag{5}$$

Each term swaps amplitude between $|01\rangle$ and $|10\rangle$, so a one-hot group stays one-hot [7]. The start state $|W_3\rangle^{\otimes G}$, with $|W_3\rangle = (|100\rangle + |010\rangle + |001\rangle)/\sqrt{3}$, is the uniform superposition of the 3^G one-hot plans [22]. The one-hot penalty is then constant and dropped; the three ring terms are applied in a fixed order. Each XY rotation compiles to two CNOTs [23], so a layer has 28 CNOTs, the same count as the X variant. We prepare $|W_3\rangle$ with an R_Y rotation, a controlled Hadamard (one CNOT) and two CNOTs, which we checked numerically; this adds 12 CNOTs once.

Grover mixer. $B_{\mathrm{GM}} = |\psi_0\rangle\langle\psi_0|$ with the same start state, so $e^{-i\beta B_{\mathrm{GM}}} = I + (e^{-i\beta} - 1)|\psi_0\rangle\langle\psi_0|$ [9]. Each layer needs the inverse preparation, a phase on $|0\rangle^{\otimes n}$ controlled by all n qubits and the preparation. With the textbook ancilla construction [24], the controlled phase uses $2(n-2)$ Toffoli gates of six CNOTs each, one controlled phase and $n - 2 = 10$ ancilla qubits, 122 CNOTs in all, so a Grover layer has 150 CNOTs.

Feasible-subspace mixer. As a reference, a Grover mixer starts from the uniform superposition of the fully feasible plans [9], [12]. Preparing it requires the feasible set, so it is a bound, not a practical circuit.

Objectives and training. We minimise either $\langle\hat{E}\rangle/\Delta$ or the CVaR with $\alpha = 0.1$ [16]. Sorting bitstrings by energy, $E_{(1)} \le E_{(2)} \le \ldots$, with probabilities $\pi_{(k)}$ and cumulative mass $\Pi_k = \sum_{j\le k} \pi_{(j)}$,

$$\mathrm{CVaR}_\alpha = \frac{1}{\alpha} \sum_k E_{(k)}\, \min\big(\pi_{(k)},\ \max(0,\ \alpha - \Pi_{k-1})\big). \tag{6}$$

Every variant is trained from scratch with the same procedure: COBYLA [25] with at most 250 evaluations per start, depth by depth. At each depth p we start from two random points ($\gamma_k \sim U(0, \pi)$, $\beta_k \sim U(0, \pi/2)$) and, for $p > 1$, from the

optimum of depth $p-1$ interpolated to p layers [15], and keep the start with the lowest objective. Training never uses the optimal plan.

A. Angle Transfer

Since the scaled Hamiltonians of all instances share the energy range $[0,1]$, we test whether trained angles carry over. Let $\Theta_p = \{\boldsymbol{\theta}_j^{(p)} : j \in \mathcal{D}\}$ be the angles trained on the development instances $\mathcal{D}$. We fix one set $\boldsymbol{\theta}_p^\star \in \Theta_p$ by one of three rules:

$$\boldsymbol{\theta}_p^\star = \arg\max_{\boldsymbol{\theta}\in\Theta_p} \ \tfrac{1}{|\mathcal{D}|}\textstyle\sum_{j\in\mathcal{D}} P_{\mathrm{opt}}(j;\boldsymbol{\theta}), \tag{7}$$

or the set with the lowest mean of $\langle\hat{E}\rangle/\Delta$, or with the lowest mean CVaR, over $\mathcal{D}$. Rule (7) needs the exact optimum of every development instance, which we obtain by exhaustive search; it is an oracle on the development set, although it uses nothing from the instances it is applied to. The other two rules use only quantities that a device can estimate. We choose among whole trained sets because their coordinate-wise median, which mixes different basins of the periodic angle space, gave $P_{\mathrm{opt}} = 0.001$ in a leave-one-out test.

V. Experimental Setup

Instances. The development set has 30 instances from twin seeds $42 + 17i$, $i = 0,\dots,29$, with the first four groups ($G = 4$, $n = 12$). All are feasible, with on average 45.0 feasible plans among the 81 one-hot plans, and in 12 of them capacity changes the optimal plan (*capacity-binding*). The held-out set has 30 further seeds, $i = 30,\dots,59$, of the same disruption pattern, used only for the final transfer test. We also built 60 instances (50 feasible, 22 capacity-binding) on the OpenStreetMap road graphs of 6 flood-prone Indian cities, with each city's mapped zones, shelters and road flood-exposure proxy. Demand is not mapped, so group sizes and seats follow the twin; the disruption closes the most flood-exposed road on any zone–shelter path and halves the first shelter's seats.

Simulation. The simulator stores all 2^n amplitudes in double precision; all random choices use fixed seeds.

Metrics. P_{opt} is the probability of measuring an optimal plan and P_{feas} that of a feasible plan. A sampler draws N plans and returns the cheapest one that passes an exact feasibility check. It returns the optimum whenever the optimum is among the samples, which has probability success@N $= 1-(1-P_{\mathrm{opt}})^N$, assuming independent shots. We also report the expected relative gap $(c-c^\star)/c^\star$ of the returned plan, given that a feasible plan was drawn, for $N \in \{8, 32, 128\}$, computed exactly from the output distribution. One evaluation is one estimate of the training objective, which on hardware is a batch of circuit executions.

Statistics. Paired comparisons use the two-sided Wilcoxon signed-rank test on per-instance values, also for success@N. We apply the Holm correction within each family of tests: the four depths of one mixer comparison, the 7 fixed-versus-trained comparisons of Table II, and the four city comparisons.

TABLE I
Mixers and objectives at depth 4 (30 development instances, 12 qubits)

Variant	P_{opt}	95% CI	P_{feas}	s@32 %	Evals	CNOT
X, mean	0.040	0.028–0.055	0.569	60.8	2,091	28
XY, mean	0.188	0.127–0.257	0.755	82.5	2,510	28
XY, CVaR	0.139	0.117–0.166	0.664	97.7	1,356	28
Grover, mean	0.139	0.121–0.157	0.815	96.9	2,033	150
Grover, CVaR	0.117	0.106–0.127	0.563	96.2	918	150
Feasible subspace	0.444	0.385–0.511	1.000	100.0	2,039	n/a

X, XY: mean over 5 seeds, CI by two-level bootstrap. Evals: objective evaluations per instance, depths 1–4. CNOT: per layer, all-to-all connectivity.

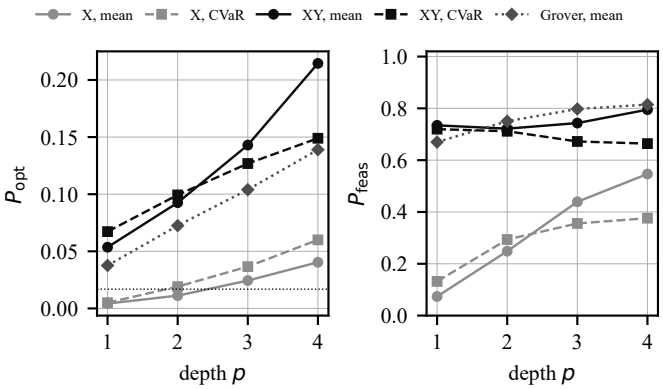


Fig. 1. Probability of the optimal plan (left) and of any feasible plan (right) against depth, optimizer seed 0. The two panels have different y-axis ranges. The dotted line is uniform sampling of one-hot plans.

Effect sizes are median paired differences. Intervals for variants run with 5 optimizer seeds use a two-level bootstrap that resamples instances and then one seed per instance. Because the city instances cluster by city, we also apply a sign test to the six per-city mean differences.

Baselines. Uniform sampling over the 81 one-hot plans ($P_{\mathrm{opt}} = 0.0169$) and over the feasible plans ($P_{\mathrm{opt}} = 0.049$); exhaustive search; a greedy rule that places the largest group first in its cheapest shelter with seats left; and simulated annealing (SA) [26] with 20 restarts of 200 sweeps, either with bit flips on (3) or with moves that reassign one group, which keep the plan one-hot. For the X mixer we swept the penalty $\lambda\mu$ with $\lambda \in \{0.25, 0.5, 1, 2\}$.

VI. Results

A. Mixers, Objectives and Penalties

Table I and Fig. 1 give the comparison. Over 5 seeds the XY mixer reached $P_{\mathrm{opt}} = 0.188 \pm 0.020$ at $p = 4$ (seed means 0.168–0.215), against 0.035–0.048 for the X mixer. It was better on 25–29 of 30 instances at every depth (Holm-adjusted $p \le 8.0 \times 10^{-6}$), and the ranking held for every seed. Its feasibility stayed between 0.722 and 0.795 at every depth, whereas the X mixer began with $P_{\mathrm{feas}} = 0.074$ and spent its layers recovering feasibility.

The Grover mixer reached 0.139 at $p = 4$, and its difference from XY was not significant at any depth (Holm-adjusted

$p \geq 0.19$). The data lean towards XY: at $p = 4$ it was ahead on 20 of 30 instances, with a mean difference of 0.076 (paired bootstrap interval 0.018–0.141) and an unadjusted $p = 0.047$. The Grover mixer spread probability more evenly over instances, so its success@32 was higher, but its layer costs 150 CNOTs and 10 ancillas against 28 CNOTs for ring-XY.

CVaR training lowered the XY mixer's mean P_{opt} but raised success@32 from 88.4% to 98.1% (seed 0): the worst instance rose from 0.002 to 0.074. The worst instances were all capacity-binding.

We tested whether a smaller penalty helps the X mixer: lowering the weight to $\lambda = 0.25$ raised its P_{opt} by 2.6× to 0.104 at a feasibility of only 0.401, still about half of the XY result. To bound the cost of keeping capacity as a penalty, we ran a mixer restricted to feasible plans: it reached $P_{\text{opt}} = 0.444$ at $p = 4$, 9.0 times uniform sampling over the same feasible plans, and beat XY on 26 of 30 instances ($p = 4.4 \times 10^{-6}$).

B. Angle Transfer to Held-Out Seeds

Table II gives the held-out test. For the XY mixer with the mean objective, the set chosen by (7) reached $P_{\text{opt}} = 0.306$ against 0.163 for per-instance training with 2,496 evaluations (better on 24 of 30, median difference 0.195, Holm $p = 8.5 \times 10^{-4}$). The oracle-free sets also beat per-instance training on these seeds: 0.264 for the energy rule ($p = 3.8 \times 10^{-3}$) and 0.234 for the CVaR rule ($p = 0.039$). The leave-one-out test on the development set tells a different story. There, the energy rule gave 0.082 and the CVaR rule 0.164, both below the 0.215 of per-instance training, while rule (7) gave 0.292. The same held for CVaR angles chosen by CVaR (0.250 on held-out seeds, 0.095 in leave-one-out against 0.149). An oracle-free choice therefore did not give a reliable gain. Rule (7) gained in both tests; the optima it needs come offline from exhaustive search or integer programming at this size. Transfer did not rescue the X mixer (0.036).

Part of the gap is a mismatch of objectives. Per-instance training minimises energy well: its mean $\langle \hat{E} \rangle / \Delta$ was 0.300 against 0.530 for the fixed set, lower on 24 of 30 instances. Low energy did not imply a high P_{opt}, since across the 30 trained sets the rank correlation between mean energy and mean P_{opt} on the development set was only 0.33. Stronger optimizers stayed below: the best of five COBYLA seeds by training objective reached 0.203 with 12,485 evaluations per instance, and L-BFGS-B 0.225. COBYLA started from the fixed angles did improve on them (0.385 with 814 evaluations, $p = 1.9 \times 10^{-3}$).

Why does a fixed set work? The performance of trained sets concentrates across instances: a set's mean P_{opt} on the development set predicted its mean on the held-out seeds with rank correlation 0.98. Only 7 of the 30 trained sets beat per-instance training there (median set 0.062), so the choice matters. The gain came mostly from capacity-binding instances: on the 16 binding seeds per-instance training reached 0.106 and the fixed set 0.312, on the others 0.228 and 0.298. We repeated training, selection and the held-out test without the scaling and with division by the standard deviation of the penalty-free plan energies. Without scaling the fixed angles reached only 0.029, although per-instance training still reached 0.270; with the other normalisation they reached 0.282, close to our 0.306. Transfer needs a common energy scale, not our particular one. The one-time cost was 75,392 evaluations of training and 3,600 for selection, the cost of per-instance training on about 32 disruptions.

C. Sampling with a Feasibility Check

Table III gives what the sampler of Section V returns. With 8 samples, the fixed XY angles came within 0.09% of the optimal cost on average, per-instance training within 1.49% and uniform one-hot sampling within 6.7%. With 32 samples every XY variant drew a feasible plan with probability 100.0% to one decimal, and with 128 samples per-instance training was within 0.10%. Many plans cost close to the optimum, so success@N is the stricter measure.

D. Transfer to Real Road Networks

Table IV breaks the city test down by city, with the angles fixed on the twin. Pooled over the 50 instances, the set chosen by (7) reached $P_{\text{opt}} = 0.202$ against 0.213 for per-instance training (Holm $p = 0.74$) and the energy-rule set 0.216 ($p = 0.38$). The fixed set led in only 3 of 6 cities (sign test $p = 1.00$): no city-level gain in P_{opt}. Per-instance training left 8 instances with $P_{\text{opt}} < 0.01$, the fixed set 1, and success@32 rose from 74.9% to 89.9% (Holm $p = 0.042$) and to 96.7% for the energy rule ($p = 9.7 \times 10^{-5}$), in all 6 cities (sign test $p = 0.03$).

E. Larger Instances and Gate Noise

We applied the 12-qubit angles of rule (7) unchanged to 10 new twin instances with five groups (15 qubits) and 10 with six groups (18 qubits). They kept P_{opt} at least 0.176 at 15 qubits and 0.056 at 18 qubits, 36 and 34 times the mean of uniform one-hot sampling, while per-instance training fell below 0.01 on 3 and 6 instances. In mean P_{opt}, at 15 qubits the fixed angles were higher on average but not significantly so ($p = 0.105$, Wilcoxon); at 18 qubits the fixed angles were higher on average but not significantly so ($p = 0.557$, Wilcoxon). With ten instances per size, the test shows that fixed angles remain usable at 15 and 18 qubits; it does not establish a gain.

We simulated gate noise by quantum trajectories (200 runs per circuit) on the first 10 development instances: every two-qubit gate, including four gates per W_3 preparation, fails with probability ϵ and then applies one of the 15 non-identity two-qubit Pauli operators. Each ZZ and XY gate compiles to two CNOTs, so $\epsilon = 0.01$ corresponds to a CNOT error near 0.5%, of the order of the 0.36% (isolated) and 0.62% (simultaneous) two-qubit Pauli errors reported for a superconducting processor [27]. At $\epsilon = 0.01$ and $p = 4$ the XY mixer kept $P_{\text{opt}} = 0.105$, 3.2× the X mixer's 0.033. A global depolarising

TABLE II

ANGLES AT $p = 4$ ON THE 30 HELD-OUT SEEDS: PER-INSTANCE TRAINING AGAINST ONE FIXED ANGLE SET CHOSEN ON THE DEVELOPMENT SET

Variant	Angles	Needs optima	Held-out P_{opt}	Median paired diff.	Wins of 30	Holm p	Leave-one-out P_{opt} (dev.)	s@32 %
XY, mean	per instance	no	0.163				0.215	79.7
	best of 5, min. energy	no	0.203				0.245	84.9
	L-BFGS-B	no	0.225					76.9
	fixed, max. P_{opt}	yes	0.306	+0.195	24	8.5×10^{-4}	0.292	100.0
	fixed, min. energy	no	0.264	+0.131	24	3.8×10^{-3}	0.082	99.9
	fixed, min. CVaR	no	0.234	+0.102	23	0.039	0.164	99.8
XY, CVaR	per instance	no	0.138				0.149	97.9
	fixed, max. P_{opt}	yes	0.250	+0.070	29	6.5×10^{-8}	0.259	99.8
	fixed, min. CVaR	no	0.250	+0.070	29	6.5×10^{-8}	0.095	99.8
X, mean	per instance	no	0.028				0.040	50.3
	fixed, max. P_{opt}	yes	0.036	+0.010	20	0.039	0.039	63.6
	fixed, min. energy	no	0.034	+0.006	21	0.064	0.034	65.8

Fixed sets vs. per-instance training of the same variant (Wilcoxon signed-rank, Holm over 7 rows). Per-instance rows: development-set mean in the leave-one-out column.

TABLE III

BEST VERIFIED PLAN AMONG N SAMPLES (HELD-OUT SEEDS, $p = 4$): MEAN GAP TO THE OPTIMAL COST AND CHANCE OF DRAWING THE OPTIMUM, IN %

	$N = 8$		$N = 32$		$N = 128$	
Sampler	gap	opt.	gap	opt.	gap	opt.
uniform one-hot	6.71	14.1	2.17	44.7	0.22	88.3
uniform feasible	4.17	26.8	0.96	67.7	0.05	96.4
XY, per instance	1.49	55.0	0.48	79.7	0.10	92.8
XY, fixed (max. P_{opt})	0.09	93.9	0.00	100.0	0.00	100.0
XY, fixed (min. energy)	0.22	87.7	0.00	99.9	0.00	100.0

TABLE IV

OPENSTREETMAP CITY INSTANCES, XY MIXER, $p = 4$: MEAN P_{opt} AND SUCCESS@32 (%)

		P_{opt}			s@32 %		
City	inst.	trained	fix-O	fix-E	trained	fix-O	fix-E
Delhi	10	0.142	0.184	0.164	76	91	94
Chennai	10	0.137	0.165	0.215	66	92	96
Mumbai	10	0.212	0.258	0.248	95	100	100
Kolkata	5	0.341	0.220	0.247	64	84	97
Guwahati	10	0.229	0.155	0.207	75	75	96
Patna	5	0.354	0.274	0.248	61	99	99
All	50	0.213	0.202	0.216	75	90	97

fix-O: set chosen by max. mean P_{opt} (7); fix-E: by min. mean energy.

model, in which a circuit with N_2 two-qubit gates returns its ideal state with probability $F = (1 - \epsilon)^{N_2}$,

$$P_{\text{opt}}^{\epsilon} = F\, P_{\text{opt}} + (1 - F)\, |\mathcal{X}^{\star}|/2^n, \tag{8}$$

with $\mathcal{X}^{\star}$ the optimal bitstrings, gave a lower 0.086, since many Pauli errors barely change the output.

F. Encoding Exactness and Classical Baselines

In all 30 development instances some infeasible one-hot plan had zero penalty, because three groups whose pairs fit overloaded a shelter together. Such plans usually cost more than the optimum, so the minimiser of (3) was the true optimum in 93.3% of instances. Exhaustive search took 0.90 ms per instance. The greedy rule found a feasible plan in 100.0% of instances and the optimum in 93.3%. Bit-flip SA ended at the optimum in 80.0% of instances and SA with one-hot moves in 93.3%, the limit set by the encoding; both visited the optimum every time, so keeping the cheapest verified plan found it in 100%.

VII. DISCUSSION AND LIMITATIONS

Three findings are relevant for hardware experiments. First, the one-hot constraint belongs in the mixer: at the same CNOT count, ring-XY beat the transverse-field mixer on almost every instance and seed, and a smaller penalty did not close the gap. Capacity remains open: the feasible-subspace mixer roughly doubled P_{opt} but needs the feasible set. Second, transfer needs a common energy scale and a good selection rule. With development-set optima, one fixed set per depth matched or exceeded per-instance training at 12 qubits after a one-time offline cost of 78,992 evaluations; without optima the gain was not reliable, and part of it reflects that energy is a weak proxy for P_{opt}. Third, the encoding admits infeasible penalty-free plans, so a verifier must check every sample.

The study has clear limits. All held-out twin instances share one disruption pattern; other closures, failed shelters and capacity cuts were not tested, and the city instances use real roads but sampled demand. The size test stops at 18 qubits with ten instances per size; simulating the circuits on the 3^G one-hot amplitudes would allow more. The noise model charges errors per logical gate rather than per CNOT, ignores single-qubit and correlated errors and omits the Grover mixer, and no circuit was compiled for hardware. Absolute P_{opt} values depend on the training budget, and the Grover comparison cannot show equivalence. Future work will test varied disruptions, slack-variable encodings and hardware runs.

VIII. CONCLUSION

On flood shelter-assignment instances, a ring-XY mixer gave QAOA a significantly higher probability of the optimal plan than the transverse-field mixer at the same CNOT count, while capacity, kept as an approximate pairwise penalty, requires a verifier. Angles fixed by mean optimal-plan probability on development instances reached $P_{\text{opt}} = 0.306$ on held-

out seeds of the same disruption pattern, above per-instance training; rules without optima were not reliable, and transfer failed without a shared energy scale. Exhaustive search solves these instances in about a millisecond, so the results guide choices for hardware experiments and show no present-day advantage.

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